

## Sungmin Kim

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**SUMMARY:** AI Computer Vision Research internship at SPAID.  
Experience in Computer Graphics, Computer Vision, Game Development and Web Development.  
Creative Student with High Quality Ideas.

**EDUCATION:** **Ajou University College of Computing and Informatics**, Suwon, Republic of Korea  
**Major: Digital Media & Applied Artificial Intelligence, (expected to graduate at Feb 2027)**  
**GPA: 3.71/4.5 (재수강 진행중인 과목의 이전 성적을 제외한 GPA)**  
**Courses taken included:**  
**Computer related Courses:** Object Oriented Programming, Data Structure, Algorithm, Computer System, Computer Graphics, Digital Image Processing, Computer Vision, Practical Web Development, GPU Programming, Introduction to AI, Machine Learning, Animation Theory  
**Math:** Probability and Statistics, Discrete Math, Calculus, Linear Algebra(SNU Summer session)

**EXTRACURRICULAR EDUCATION:** **Stanford Online:** CS231N Deep Learning for Computer Vision  
**AWS Skill Builder:** Amazon Bedrock Getting Started, AWS Technical Essentials, Cloud Essentials – Knowledge Badge Readiness Path  
**NXT Cloud:** Ajou University Sprout Camp (AWS)

**EXPERIENCE:** **Private Tutor, Mar 2021 – Dec 2022 & Aug 2024 – Present**  
**Yonsei Naro Academy, Suwon & Hwaseong Campus**  
- Started with Korean and Math tutor but, most of the time worked as a Science (Integrated Science, Life Science & Chemistry) tutor for high school students  
- Provided detailed lectures and personalized problem sets to let students score higher at exams  
- Informed parents monthly about the overall plans for the class  
- Give information about various college majors based on the real experience from myself and my acquaintances

**Military Service as a Linguist (Translator), Jan 2023 – Jun 2024**  
**U.S. Army Garrison Humphreys, 501<sup>st</sup> MI Brigade, Pyeongtaek**  
- Worked as a translator in various situations from casual conversations to briefings and meetings with US and ROK army generals  
- Participated in three Combined Training and received 1 COA from ROKA, 4 COA from US Army (COA: Certificate of Achievement or Appreciation)  
- Worked as a Teaching Assistant in Korean Language class for US army soldiers

**Teaching Assistant, Mar 2025 – Jun 2025**  
**“Introduction to Bigdata and Analysis” by Professor Wonseon Song, Ajou University**  
- Helped students understand key concepts in analysis by answering questions  
- Read and scored all submitted programming assignments

**AI Research Intern, July 2025 – Aug 2025**  
**SPAID, Seoul, Republic of Korea**  
- Constructed a robust custom dataset by curating road-view images to resolve class imbalance issues in public datasets, specifically targeting modern architectural styles for

better generalization in urban environments.

- Optimized the YOLO-seg model based on Python/PyTorch, achieving improved inference performance by fine-tuning data augmentation parameter(e.g., Mosaic, Mixup) at the code level to mitigate overfitting.

- Established an automated pipeline that extracts inference results into JSON format, enabling seamless integration with 3D texture mapping processes and demonstrating end-to-end service implementation capabilities.

**AWARDS:**

**Smart Railroad Safety Idea Contest using 4<sup>th</sup> Industrial Revolution Technology, 2017**

**President of Korea Railroad Research Institute Award (2<sup>nd</sup> Place):**

“Railroad facility Monitoring System using Noise and Vibration data collected from Individuals’ Smartphones”

**COMPUTER SKILLS:**

**Python:** Solved approximately 400 algorithm questions in Baekjoon Online Judge, familiar with various packages including Pytorch

**R:** Completed introductory courses and education from various institutions

**Java:** Learned Java using Head First Java(OREILLY)

**C++:** Studied the basic syntax of C++ for OpenGL and OpenCV

**Unity:** Completed a simple 2D game by following a tutorial

**Web development:** Used React & Supabase to build a webservice for science education

**LANGUAGE SKILLS:**

**Korean (Native), English (Fluent – TOEFL 107)**

**Project:**

**Frequency-Aware Dual-Branch LIIF for Super-Resolution:**

- Proposed a data-centric method to improve the performance of the LIIF model.

Our method prepares an additional LIIF model that is trained on high-frequency residual data acquired from the original dataset and let it output its own results. Then, it is linearly combined with the inference result of the baseline LIIF model.

Our model was able to output a result containing high-frequency details. When compared with our baseline LIIF, it displayed maximum 0.12 improvement in PSNR and 0.0061 in SSIM.

**CERTIFICATES & PORTFOLIO:**

**Bigdata Analysis by KRDI (Date Acquired: August 5, 2022)**

**MITx 6.00.1x: Introduction to Computer Science and Programming Using Python (Date Acquired: March 30, 2024)**

**HEMP: Health Evaluation Metric using Proposals:** <https://imgur.com/8j8t8/1K6DTd9F0GRdQ4LEBGu4nrg>

**Cinematic Trailer for Books(1984):** <https://clips.twitch.tv/188V7HGaMwN8gVRm1mVZrRZ/iewAupshing>

**Kimchi run:** <https://play.unity.com/ko/games/a0140f4a-1717-471f-8632-f04495c27eea/kimchi-run-by-mrkim>

**Internship at SPAID:** [https://www.wondersharepdfelement.com/work/2470d867606809fae2ef8885552a?source=copy\\_link](https://www.wondersharepdfelement.com/work/2470d867606809fae2ef8885552a?source=copy_link)

**PLEARN:** <https://pwd-final.vercel.app/>